Submission Report: School Data Management System

Project Title:

School Data Management System

Overview

The School Data Management System allows users to manage student records efficiently. This system includes functionalities to add, view, search, update, and delete student data. The project demonstrates concepts like input validation, arrays, loops and conditionals.

Features and Functionalities

Welcome Screen (welcome())

Displays an introductory welcome message in ASCII art style.

```
__        __     _
\ \      / /__   | | _______  ________   ___
 \ \ /\ / / __ \ | |/ __/ _ \| '_ ` _ \ / _ \
  \ V  V / /__/  | ||(__ (_) | | | | | |  __/
   \_/\_/\____   |_|\___\___/|_| |_|_|_|\___|

---------------------------------------------
|  SCHOOL DATA MANAGEMENT SYSTEM OF STUDENT  |
---------------------------------------------

NOTE: First You have to Provide/Enter Data of
      Student Otherwise Project won't' work

---------------------------------------------

Press 1 to Enter Data
Press 2 to Show Data
Press 3 to Search Data
Press 4 to Update Data
Press 5 to Delete Data
Press 6 to Exit
```

Purpose: To create an engaging start to the application.

Main Menu

Displays options for different functionalities:
1. Enter Data
2. Show Data
3. Search Data
4. Update Data
5. Delete Data
6. Exit

Purpose: To allow users to navigate through different operations.

Input validation ensures users only enter valid menu options (1-6).

Enter Data (enter())

Collects student details like name, roll number, course, class, and contact number.

Key Points:

- - Ensures only positive integers are accepted for the number of students.
- - Uses getline() to handle spaces in input fields like names and courses.
- - Updates the total number of students (total).

Validation: Checks input validity and handles input errors.

```
--------------------------------
    Enter data of Student 1     |
--------------------------------
Enter Name    : sheeraz ahmed
Enter Roll No.: 1234
Enter Course  : programming
Enter Class   : Bscs
Enter Contact : 03318940245

--------------------------------
    Enter data of Student 2     |
--------------------------------
Enter Name    : shahzeb ali
Enter Roll No.: 1235
Enter Course  : programming
Enter Class   : Bscs
Enter Contact : 03111179839

--------------------------------
```

Show Data (show())

Displays the stored student data in a formatted style.

Key Points:

- - Iterates over the student array and prints the details.
- - Provides a visual separator for readability using dashes (-).

```
*********DATA OF REGISTERED STUDENTS*********
---------------------------------
       Data of Student 1        |
---------------------------------
Name    : sheeraz ahmed
Roll No.: 1234
Course  : programming
Class   : bscs
Contact : 0331890245
---------------------------------
---------------------------------
       Data of Student 2        |
---------------------------------
Name    : shahzeb ali
Roll No.: 1235
Course  : programmming
Class   : bscs
Contact : 03111179839
---------------------------------
```

Search Data (search())

Searches for a student using their roll number.

Key Points:

- - Prompts the user to enter the roll number.
- - If the roll number matches a record, displays the student's details.
- - If no match is found, notifies the user.

```
Enter Roll No: of Student which you want to search:
1234
-----------------------------
      Data of student 1     |
-----------------------------
Name    : sheeraz ahmed
Roll No.: 1234
Course  : programming
Class   : bscs
Contact : 0331890245

-----------------------------
Press 1 to Enter Data
Press 2 to Show Data
Press 3 to Search Data
Press 4 to Update Data
Press 5 to Delete Data
Press 6 to Exit
```

Update Data (update())

Updates an existing student record based on the roll number.

Key Points:

- - Displays the existing record before updating it.
- - Prompts the user to enter new details.
- - Handles cases where the roll number is not found.

```
The previous data of Student 2
-----------------------------------
        Data of Student 2         |
-----------------------------------
Name    : shahzeb ali
Roll No.: 1235
Course  : programmming
Class   : bscs
Contact : 03111179839

-----------------------------------
Enter the new Data of Student 2   |
-----------------------------------
Enter Name    : shahzeb ahmed
Enter Roll No.: 1235
Enter Course  : discrete
Enter Class   : bscs
Enter Contact : 03111179839
```

UPDATED DATA:

```
*********DATA OF REGISTERED STUDENTS*********
---------------------------------
       Data of Student 1        |
---------------------------------
Name    : sheeraz ahmed
Roll No.: 1234
Course  : programming
Class   : bscs
Contact : 0331890245
---------------------------------
---------------------------------
       Data of Student 2        |
---------------------------------
Name    : shahzeb ahmed
Roll No.: 1235
Course  : discrete
Class   : bscs
Contact : 03111179839
---------------------------------
```

Delete Data (delet())

Deletes student records. Offers two options:

1. Delete all records.
2. Delete a specific record based on the roll number.

Key Points:

- - For specific deletions, shifts the subsequent records to maintain the array structure.

- - Updates the total variable to reflect the number of remaining records.

```
Press 1 to Delete whole record:
Press 2 to Delete specific record:
2
Enter Roll No: which you want to Delete : 1235

Your required report is Deleted
```

DATA AFTER DELETION:

```
*********DATA OF REGISTERED STUDENTS*********
--------------------------------
       Data of Student 1        |
--------------------------------
Name    : sheeraz ahmed
Roll No.: 1234
Course  : programming
Class   : bscs
Contact : 0331890245
--------------------------------
```

Program Flow

1. The program begins with the welcome screen.
2. The main menu allows the user to navigate between functionalities.
3. Input is validated to ensure the program handles errors appropriately.
4. Users can perform operations such as adding, viewing, searching, updating, and deleting records.
5. The program runs in a loop until the user chooses to exit.